

Joseph Helbing

Machine Learning Engineer + Data Scientist

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EDUCATION

University of Chicago, *M.A. Computational Social Science* [2023]
Ohio State University, *M.A. East Asian Languages and Literatures (Mandarin)* [2013]
University of Illinois at Urbana Champaign, *B.A. Political Science and China Studies* [2011]

LANGUAGES

English *Native*
Mandarin *Professionally Fluent*

SKILLS

- Machine Learning (ML)
- High Performance Computing (HPC)
- Natural Language Processing (NLP)
- Agent Based Modeling and Simulation
- Python, R, SQL
- HTML/CSS
- Web Scraping
- AWS Textract, S3, GovCloud
- Data Collection, Analysis and Visualization
- Plotly, Pyvis
- Communicating Analysis to Non-Technical Audiences

SELECTED PROJECTS

Library of Congress - Large-Scale Document Processing System

- Architected and implemented an end-to-end pipeline for extracting structured data from 500,000+ military court martial documents across all U.S. military branches
- Architected an advanced document processing pipeline integrating AWS Textract, custom bounding box algorithms, and LLM refinement to extract 60+ variables from heterogeneous military forms, with built-in quality assurance through a purpose-built GUI for sampling based human verification
- Designed and implemented SQL database architecture for efficient storage and retrieval of extracted information
- Led technical training sessions for US Digital Corps NLP working group on methodology
- <https://gist.github.com/JoeHelbing/17735d23f59efbfc894068a238d9f8df>

Equal Employment Opportunity Commission (EEOC) - ML-Based Case Analysis Support

- Developed and fine-tuned bi-directional vectorization models using BERT and modernBERT architectures for free form textual resume classification in support of case investigations

Personal Research Project - Advanced Agent-Based Modeling

- Reimagined and implemented Thomas Schelling's Segregation Agent Based Model as individualized Reinforcement Learning agents using Pytorch and Flask.
- [JoeHelbing/rl_abm_exp](https://github.com/JoeHelbing/rl_abm_exp)

EXPERIENCE

US Digital Corps at Equal Employment Opportunity Commission (EEOC) [2024 – Present]

Data Scientist (Full-Time)

- Assisting EEOC investigators on AI and ML related investigations with technical consulting, forensic source code review, and case related testing.
- Driving internal modernization of analytics resources, including standing up cloud computing and virtual machine systems for data science and analytic tasks.
- Assisting EEOC data analysts and statisticians on case investigations by training Language Model based classification systems, network analysis, and other modern data science techniques.
- Taking ownership of existing codebases of internal analytics tools—upgrading and refactoring with modern best practices to achieve large speedups, limit concurrency issues, and minimize resource footprints, while adapting them to lower cost platforms.
- Co-leader of US Digital Corps NLP working group, managing group administration, facilitating inter-agency knowledge sharing, and training on NLP methods.

Library of Congress Federal Research Division [2023 – 2024]

Data Analyst (Full-Time)

- Long form research reporting for military and federal clients, including literature review, statistical data analysis, and visualization.
- Technical lead on large-scale Natural Language Processing (NLP) project utilizing a combination of AWS GovCloud and local compute based open weights foundation models in a data extraction pipeline from legal document images.
- Contributor on dual-use technologies research reports via Chinese source material for US government client.

University of Chicago Data Science Institute

[2022 – 2022]

Research Assistant (Full-Time)

- Developed a web scraper for the Security and Exchange Commission (SEC) EDGAR API to access corporate reports.
- Utilized statistical textual matching techniques, XBRL scraping, and open-source pretrained machine learning models to create an information extraction pipeline.

Paratech Inc

[2015 – 2021]

Marketing Coordinator (Full-Time)

[2019 – 2021]

- Oversaw the update of Paratech's corporate website, managed marketing materials, and fostered dealer partnerships, focusing on tech integration and staff training.
- Launched a new WordPress website, handling design, content, and CMS.
- Developed webinar series utilizing YouTube, trained the sales team, and ran live training events in the field broadcast to customers.

Industrial Sales Manager (Full-Time)

[2017 – 2019]

- Led the development of a new industrial sales market, establishing partnerships with distributors and manufacturing representatives.
- Produced industrial, military, and maritime marketing materials, including brochures and videos, using tools like InDesign, Photoshop, and DaVinci Resolve.
- Contributed on the production floor during peak times to meet manufacturing targets.

China Regional Sales Manager (Full-Time)

[2015 – 2017]

- Overhauled Chinese operations strategically and organizationally, improving the company's position and navigating away from problematic relationships without harming client networks.
- Transitioned local contacts to direct company relationships, retaining all dealers and clients.
- Handled contract design and translation between Chinese and English, and directly negotiated partnerships through multiple visits to the country.

SiMple International Inc

[2014 – 2015]

Founder (Full-Time)

- Founded company selling international telecommunications to exchange students through partnership distribution channels.

INTO University Partnerships

[2013 – 2014]

US Recruitment Manager Beijing and North China (Full-Time)

- Worked with education consultancy network to recruit students to attend degree programs in the United States in representation of 6 US universities.
- Northern China recruitment territory based in Dalian, Liaoning China, self-planning and executing travel and events, and position responsibilities executed nearly exclusively in Mandarin Chinese.